

Daktronics All Sport CG Parser Reference

NetFront ESP32 Scoreboard Bridge - CONTROL port serial feed

Serial Configuration

Setting	Value
Purpose	Read clock, running state, scores, shots, period, and penalties
Source port	Daktronics All Sport CG CONTROL port (DB-9)
Electrical interface	RS-232 through MAX3232 level converter; do not connect DB-9 RS-232 directly to ESP GPIO
UART	9600 baud, 8 data bits, no parity, 1 stop bit (8N1)
Flow control	None
ESP32 pins	RX GPIO16, TX GPIO17

Frame Structure

<01>[fixed-width payload]<04>

Element	Hex	Meaning
Start marker	0x01	SOH; resets frame capture and starts a new payload
End marker	0x04	EOT; completes the frame and invokes parsing

All column numbers below are zero-based offsets within the payload, excluding SOH and EOT.

Main Mapped Fields

Columns	Width	Mapped field	Parser behavior
0-4	5	Game clock	MM:SS ; minutes at 0-1, colon at 2, seconds at 3-4
5-6	2	Reserved / spacing	Ignored
7	1	Clock state	s or S = stopped; blank = running
8-9	2	Home score	Integer, clamped to 0-199
10-11	2	Away score	Integer, clamped to 0-199
12	1	Reserved / spacing	Ignored
13-15	3	Home shots	Integer, clamped to 0-199
16-17	2	Away shots	Integer, clamped to 0-199
18	1	Period	Integer, clamped to 1-9
19	1	Separator	Ignored

Penalty Mapped Fields

Each penalty slot is exactly seven characters with the format `PP M:SS` .

Columns	Mapped slot	Example player	Example remaining time
20-26	Home Penalty 1	25	1:34
27-33	Home Penalty 2	36	1:38
34-40	Away Penalty 1	54	1:45
41-47	Away Penalty 2	55	1:50

Penalty Slot Subfields

Relative columns	Meaning	Validation
0-1	Player number	Must contain a player number greater than zero
2	Separator	Normally a space
3	Minutes remaining	Single digit in current observed format
4	Colon	Must be <code>:</code>
5-6	Seconds remaining	Must be 00-59

A valid slot with remaining time above `0:00` is active. A blank, malformed, or zero-time slot is cleared/inactive. The CG feed remains authoritative for penalty time in hardware mode.

Observed Example

`<01>16:33 s 2 211 1 25 1:3436 1:3854 1:4555 1:500<04>`

Mapped value	Decoded result
Clock	16:33, stopped
Home Penalty 1	Player 25, 1:34 remaining
Home Penalty 2	Player 36, 1:38 remaining
Away Penalty 1	Player 54, 1:45 remaining
Away Penalty 2	Player 55, 1:50 remaining

Unknown trailing field: payload data at column `48+` has included values such as `0` and `16` . It is likely a control or checksum field and is intentionally ignored until its definition is confirmed.

ESP Output Fields

The full state and WebSocket payload expose these mapped properties:

- clock
- clockFormatted
- clockRunning
- clockMin
- clockSec
- clockTenths
- homeScore
- awayScore
- homeShots
- awayShots
- period
- homePen1Player
- homePen1Time
- homePen2Player
- homePen2Time
- awayPen1Player
- awayPen1Time
- awayPen2Player
- awayPen2Time

Serial Analyzer Penalty Array

```
{
  "team": 0,
  "slot": 1,
  "player": 25,
  "minutes": 1,
  "seconds": 34
}
```

Value	Meaning
team: 0	Home
team: 1	Away
slot: 1	First penalty slot for that team
slot: 2	Second penalty slot for that team

Parser Behavior

- UART bytes are captured between SOH and EOT into a fixed-capacity buffer.
- Native production frames use fixed column offsets and do not rely on whitespace tokenization.
- Incomplete or malformed penalty slots are cleared rather than retaining stale values.
- Clock tenths are set to zero because the observed CG hardware feed provides whole seconds only.
- The unknown trailing control/checksum field is not validated.
- The Serial Analyzer presents escaped control bytes such as <01> and <04> .

Text Test-Feed Fallback

The firmware retains a line-oriented key/value parser for simulator and diagnostic feeds. Supported keys are:

Mapped value	Accepted keys
Clock	CLK , CLOCK , TIME
Period	PER , PERIOD
Home score	HOME , HOM , HSCORE
Away score	AWAY , AWY , ASCORE
Home shots	HSLOTS , SHOTH , HSH
Away shots	ASHOTS , SHOTA , ASH
Clock running	RUN , CLOCKRUN , RUNNING
Penalty slots	HP1 , HP2 , AP1 , AP2

Relevant ESP Endpoints

Endpoint	Purpose
/api/state	Current full gateway state
/api/state/clock	Clock and period payload
/api/state/full	Explicit full scoreboard payload
/serial.json	Legacy serial state, penalty array, RX diagnostics, and raw trace
/admin.json	Admin/debug representation of full state
/ws	Authenticated WebSocket state stream